

McGill

Reactive Flows with Cantera

ASEC onboarding project: equilibrium, diffusion flames, premixed flames

Dara Rahmat Samii · McGill University · October 22, 2026

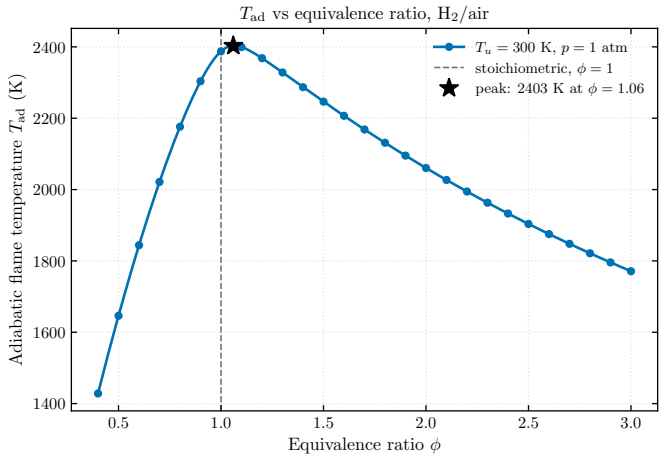
Overview

H_2/air with Cantera (`h2o2.yaml`), from thermodynamics to flames.

- **Part 1**: adiabatic flame temperature and chemical equilibrium
- **Part 2**: counterflow diffusion flames, $T(Z)$ versus scalar dissipation
- **Part 3**: freely propagating premixed flames, S_L and δ_L

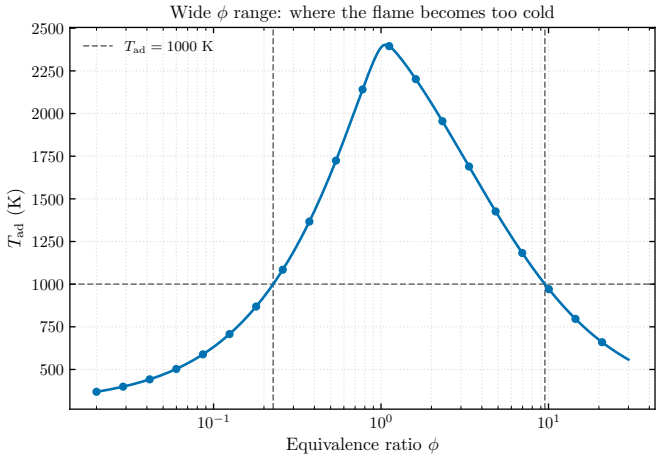
Part 1 · Equilibrium

T_{ad} vs equivalence ratio



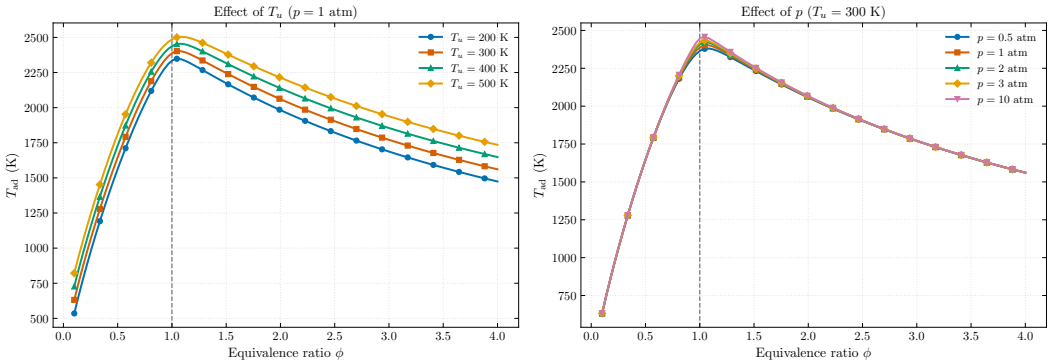
The peak (2403 K) sits at $\phi \approx 1.06$, not at stoichiometric, because slightly rich mixtures dissociate less.

Where the flame becomes too cold



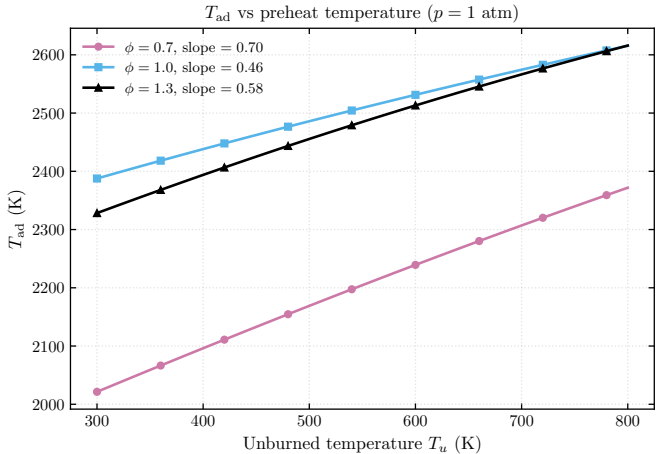
T_{ad} drops below 1000 K for $\phi < 0.23$ and $\phi > 9.5$, bracketing the flammability limits.

Effect of preheat and pressure



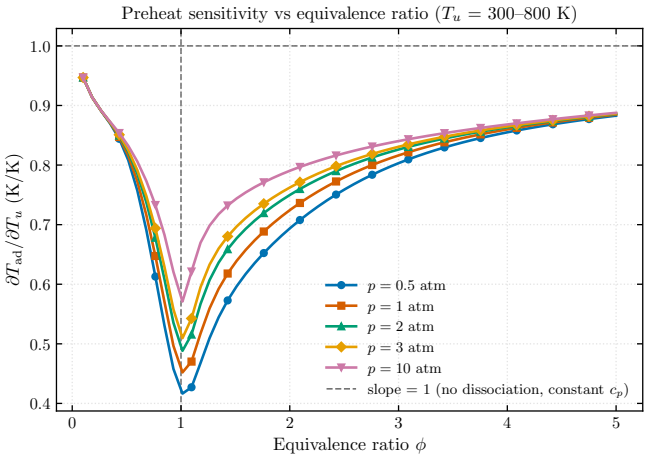
Preheat shifts the whole curve up; pressure only lifts it near $\phi = 1$, where dissociation matters.

T_{ad} vs preheat temperature



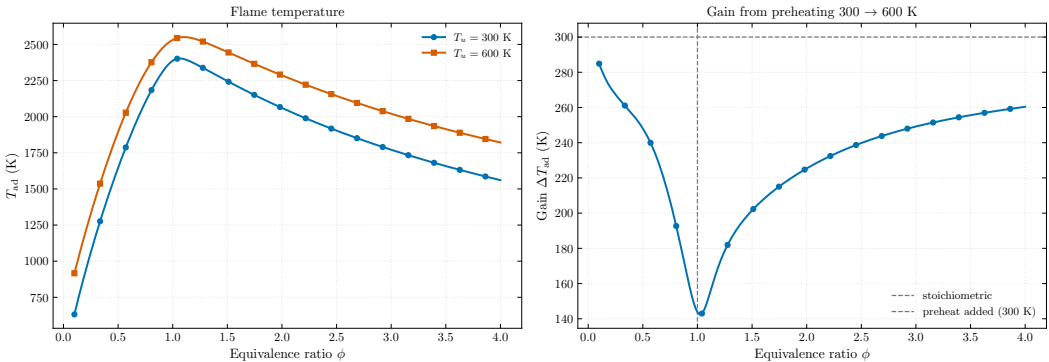
Only 0.46–0.70 K of flame temperature is gained per kelvin of preheat; the rest goes into dissociation and higher c_p.

Preheat sensitivity vs ϕ



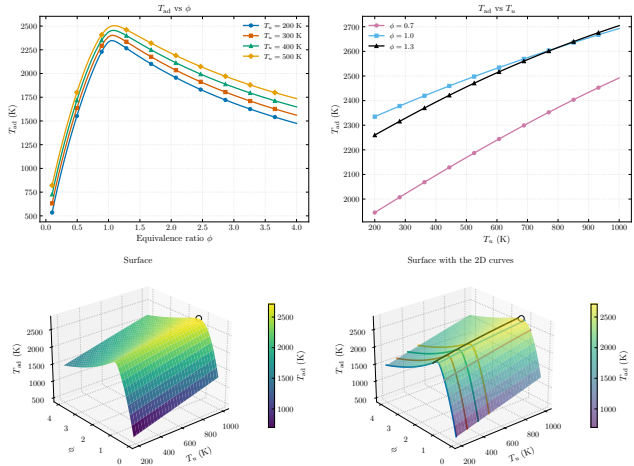
The slope is lowest at stoichiometric, where the flame is hottest and dissociates most.

Gain from preheating $300 \rightarrow 600\,\mathrm{K}$



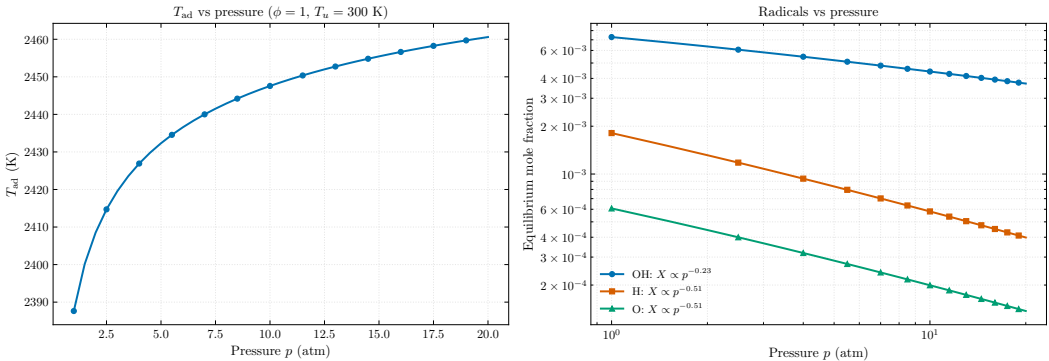
Preheating by 300 K adds only 144 K at $\phi = 1$, but up to $\sim$260 K for very lean or rich mixtures.

$T_{\mathrm{ad}}(\phi, T_u)$ **overview**



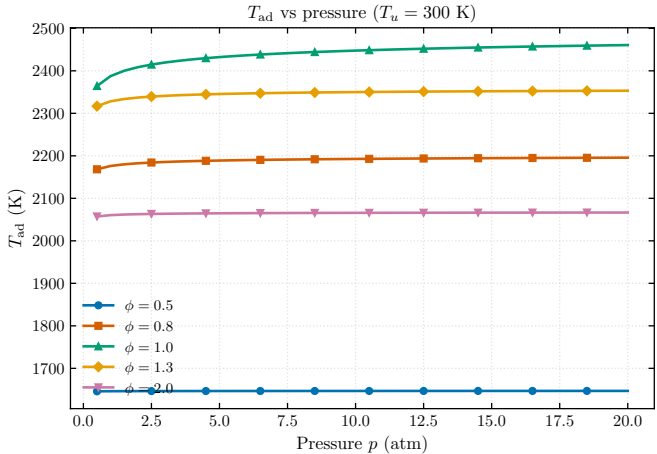
The 2D curves are cuts through one surface whose ridge stays slightly rich of stoichiometric.

Pressure and radicals



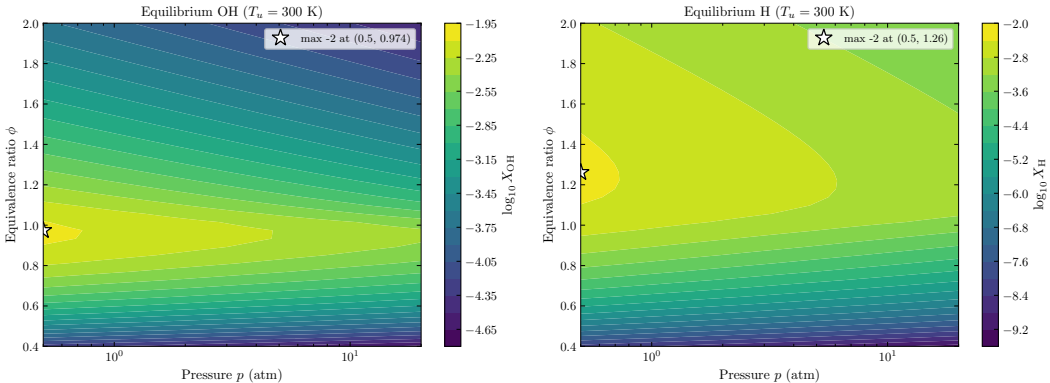
From 1 to 20 atm, T_{ad} rises only 73 K while H and O fall as $p^{-1/2}$ (Le Chatelier).

T_{ad} vs pressure for several ϕ



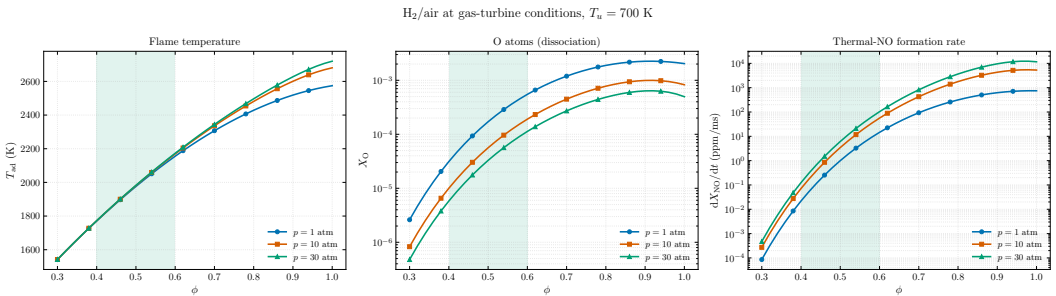
Pressure matters most near $\phi = 1$ ($+96$ K) and is negligible for the cool lean flame.

Radical maps



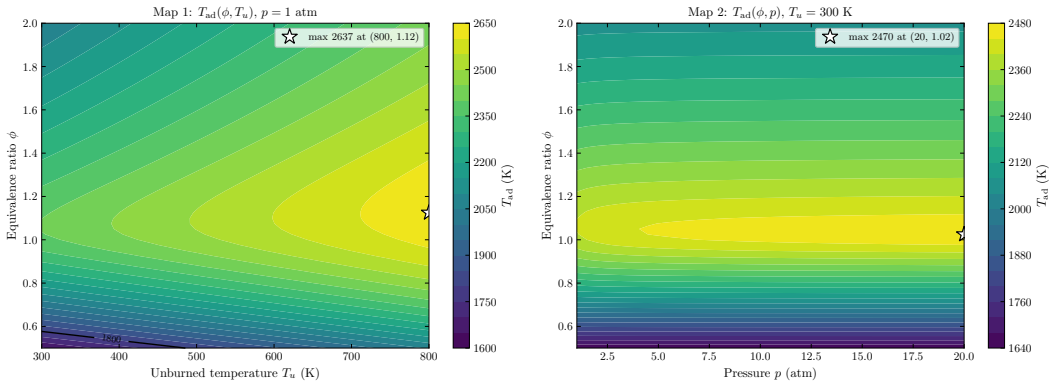
OH and H peak in a hot ridge near stoichiometric that fades as pressure rises.

Thermal NO_x



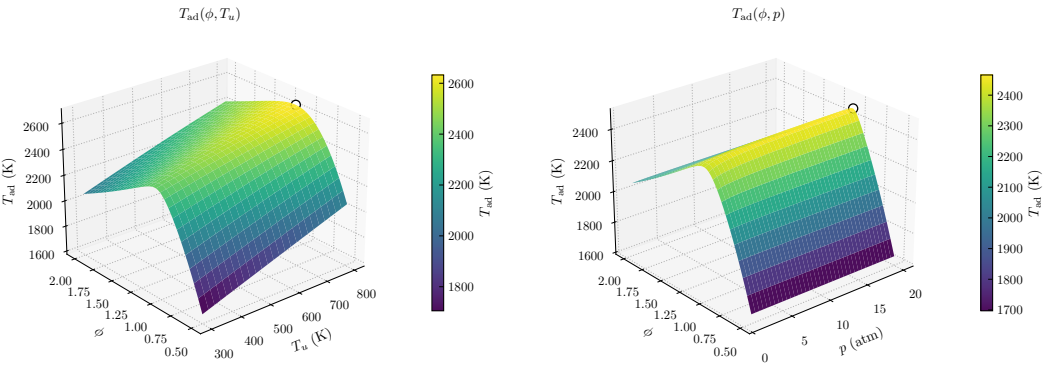
Going from $\phi = 1$ to the lean-premixed window cuts the Zeldovich NO rate by about three orders of magnitude.

2D equilibrium maps



T_u drives T_{ad} far more than p; the 1800 K isoline marks the low-NO_x lean limit.

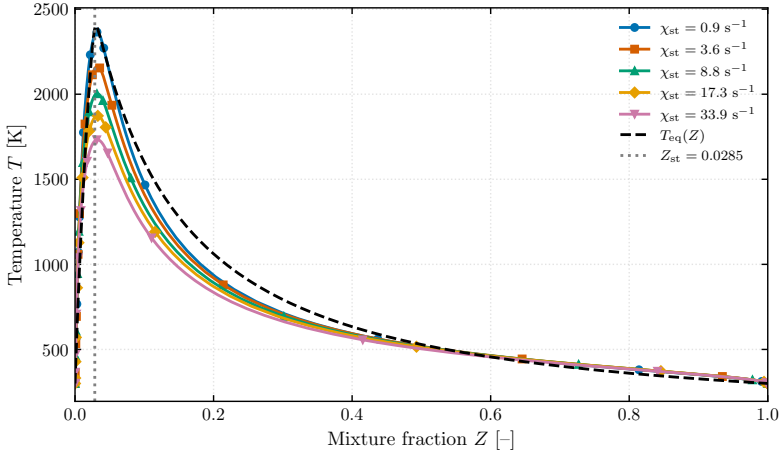
Maps as surfaces



Both maxima sit on the domain edge at slightly rich ϕ.

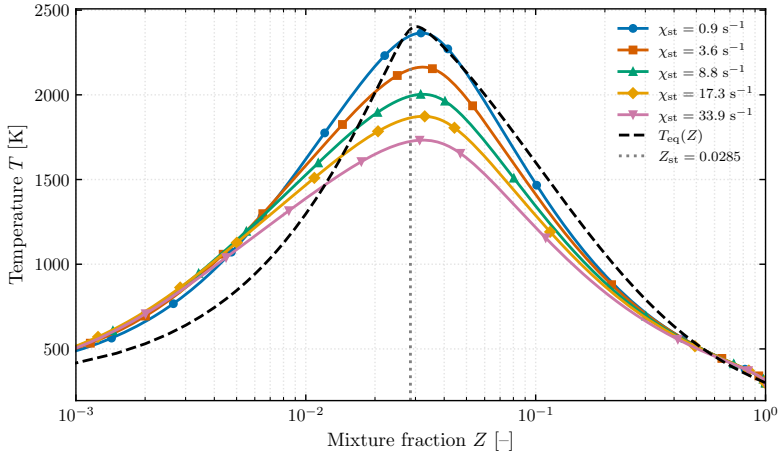
Part 2 · Diffusion Flames

$T(Z)$ at increasing scalar dissipation



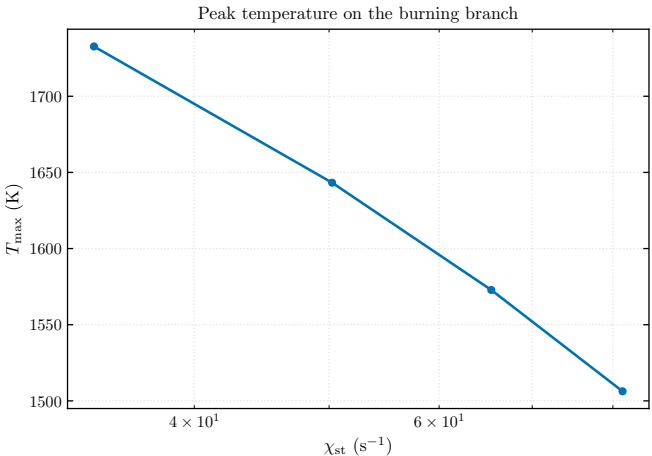
Higher χ_{st} leaves less time for chemistry, so the flame falls further below the equilibrium envelope.

$T(Z)$ on a log Z axis



On the lean side the flames exceed equilibrium, a sign of H_2 differential diffusion.

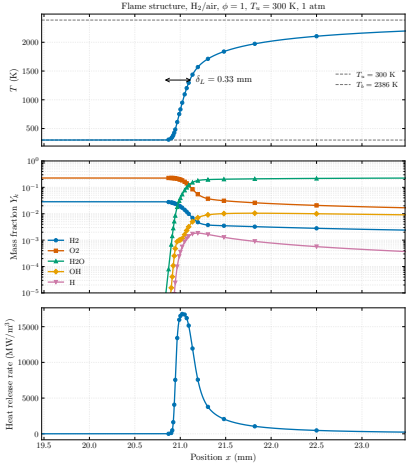
Towards extinction



The burning branch ends at $\chi_{\mathrm{st}} \approx 81\,\mathrm{s}^{-1}$, after which the solution collapses to pure mixing.

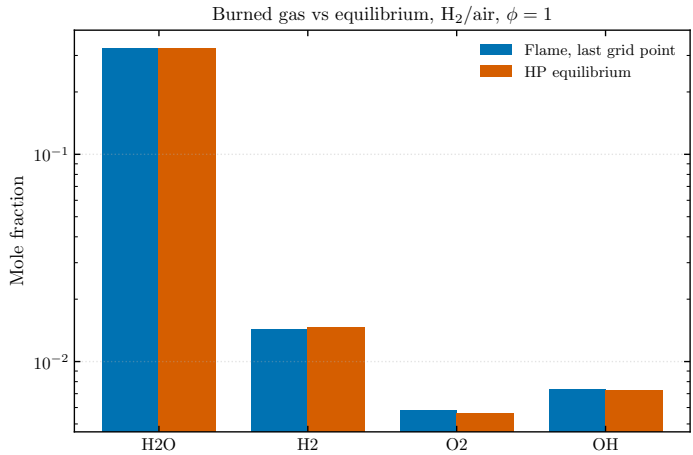
Part 3 · Premixed Flames

Flame structure



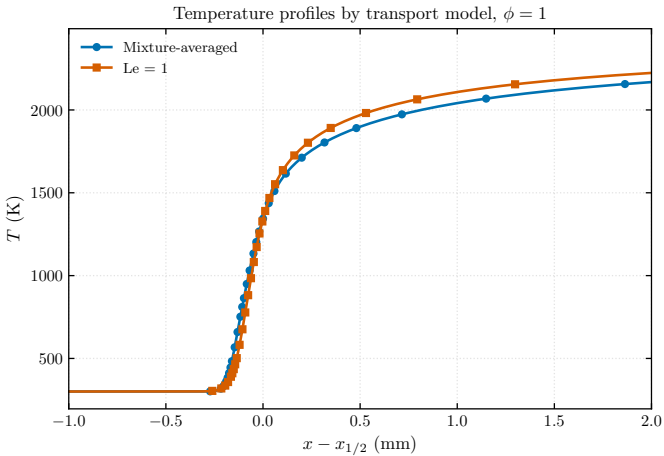
Heat release peaks at ∼1000 K, while slow recombination keeps raising the temperature downstream.

Burned gas vs equilibrium



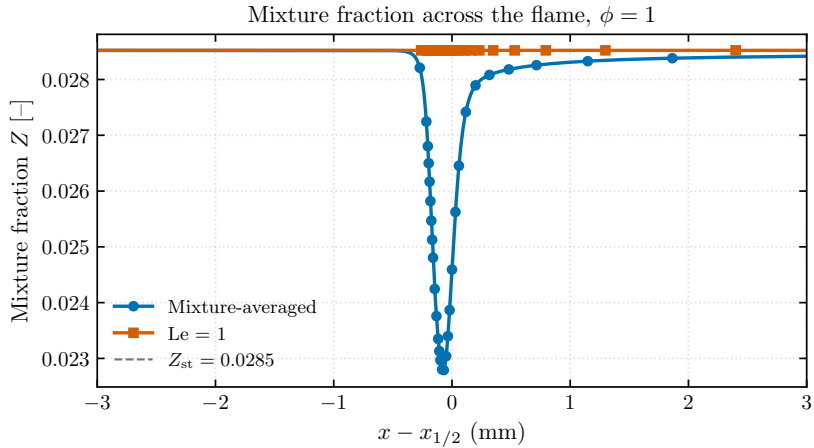
The flame's burned gas matches HP equilibrium to 0.06 % in temperature and within 4 % for each species.

Transport models: temperature



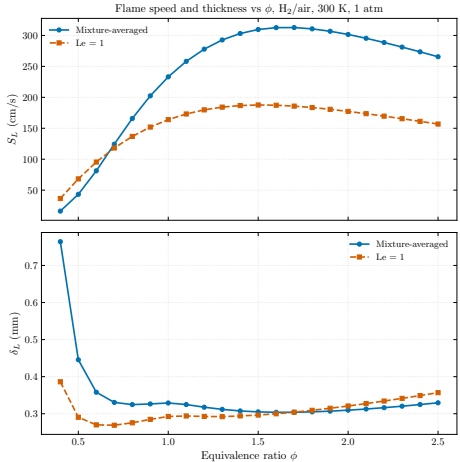
Mixture-averaged transport gives a faster flame (233 vs 164 cm/s) with a similar thickness.

Transport models: mixture fraction



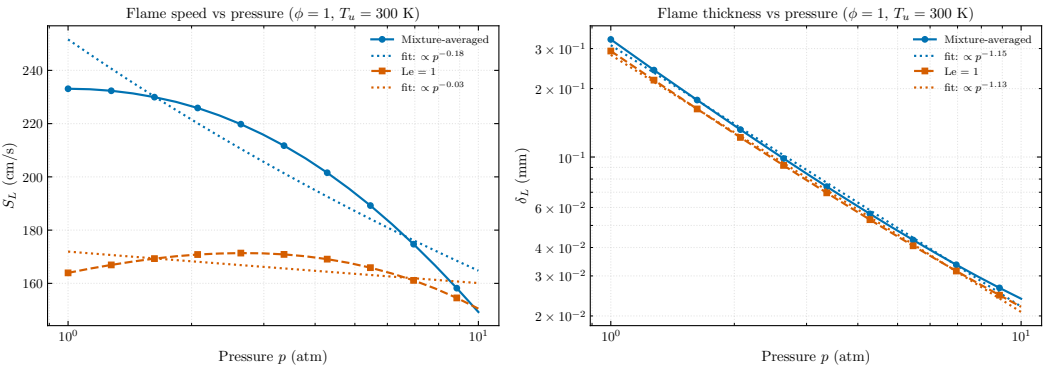
Z dips by 20 % under mixture-averaged transport because H_2 diffuses faster than heat; it stays constant for $\mathrm{Le} = 1$.

$S_L(\phi)$ and $\delta_L(\phi)$



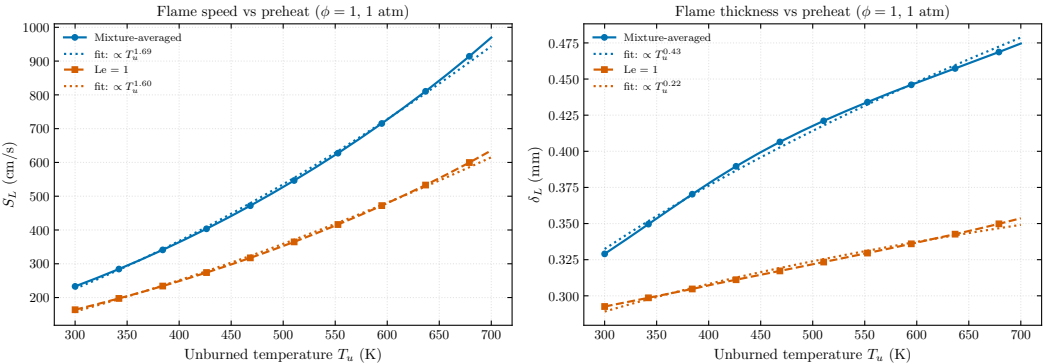
S_L peaks at $\phi \approx 1.7$, and the two transport models cross near $\phi \approx 0.67$.

Pressure sweep



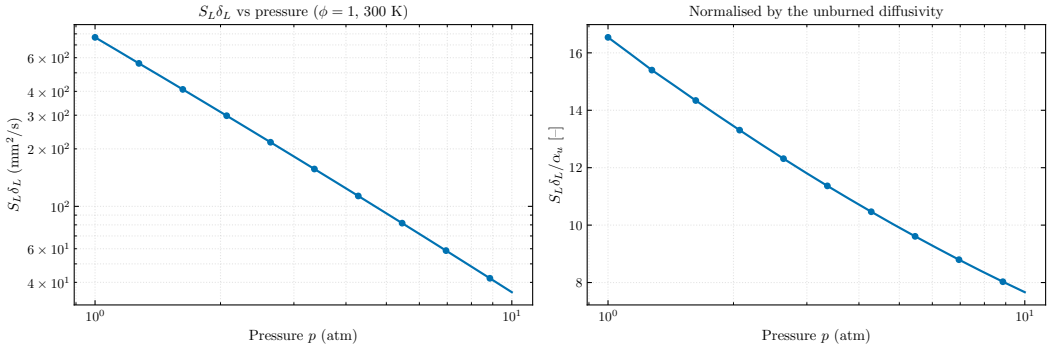
δ_L falls almost as $1/p$, while S_L drops only slowly and is not a clean power law.

Preheat sweep



$S_L \propto T_u^{1.69}$: preheating by 200 K more than doubles the flame speed.

$S_L\delta_L$ vs pressure



$S_L\delta_L$ tracks the thermal diffusivity, which scales as $1/p$; normalised by α_u it varies only twofold.